Администрирование сетевых подсистем

Лабораторная работа № 9 — Настройка POP3/IMAP сервера (Postfix + Dovecot)

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Цели и задачи работы

Цель лабораторной работы

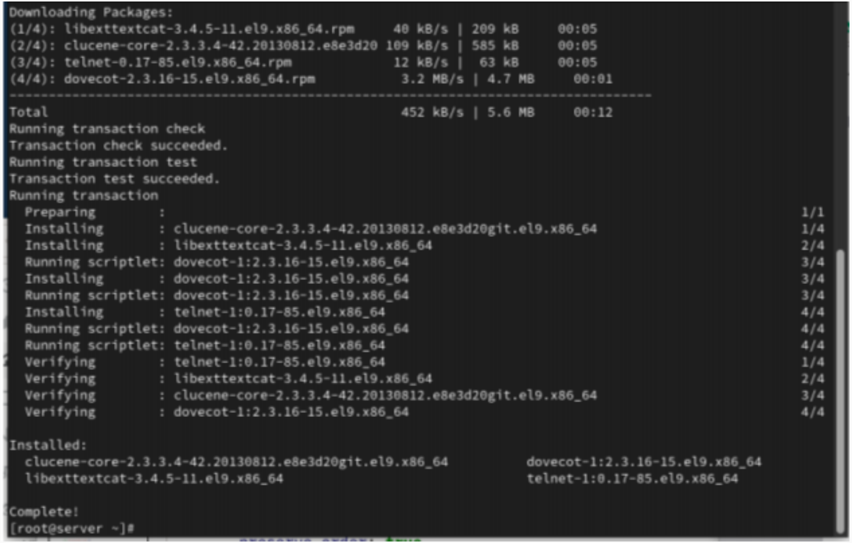
Приобрести практические навыки установки и простейшего конфигурирования POP3/IMAP-сервера.
Реализовать доступ к почте по IMAP/POP3 и проверить работу сервиса с клиентской стороны.

Выполнение лабораторной работы

Разрешённые протоколы IMAP/POP3

```
[frank@server ~]$ sudo -i
[sudo] password for frank:
[root@server ~]# dnf -y install dovecot telnet
Extra Packages for Enterprise Linux 9 - x86_64  2.3 kB/s |  38 kB     00:16
Extra Packages for Enterprise Linux 9 - x86_64  430 kB/s |  20 MB     00:47
Rocky Linux 9 - BaseOS                          392  B/s | 4.1 kB     00:10
Rocky Linux 9 - BaseOS                           77 kB/s | 2.5 MB     00:33
Rocky Linux 9 - AppStream                       191  B/s | 4.5 kB     00:24
Rocky Linux 9 - AppStream                       227 kB/s | 9.5 MB     00:42
Rocky Linux 9 - Extras                          270  B/s | 2.9 kB     00:11
Dependencies resolved.
================================================================================
 Package         Arch     Version                            Repository  Size
================================================================================
Installing:
 dovecot         x86_64   1:2.3.16-15.el9                    appstream  4.7 M
 telnet          x86_64   1:0.17-85.el9                      appstream   63 k
Installing dependencies:
 clucene-core    x86_64   2.3.3.4-42.20130812.e8e3d20git.el9     appstream  585 k
 libexttextcat   x86_64   3.4.5-11.el9                       appstream  209 k

Transaction Summary
================================================================================
Install  4 Packages

Total download size: 5.6 M
Installed size: 20 M
Downloading Packages:
(1/4): libexttextcat-3.4.5-11.el9.x86_64.rpm     40 kB/s | 209 kB     00:05
(2/4): clucene-core-2.3.3.4-42.20130812.e8e3d20 109 kB/s | 585 kB     00:05
(3/4): telnet-0.17-85.el9.x86_64.rpm             12 kB/s |  63 kB     00:05
(4/4): dovecot-2.3.16-15.el9.x86_64.rpm          3.2 MB/s | 4.7 MB     00:01
```

Проверка механизма аутентификации



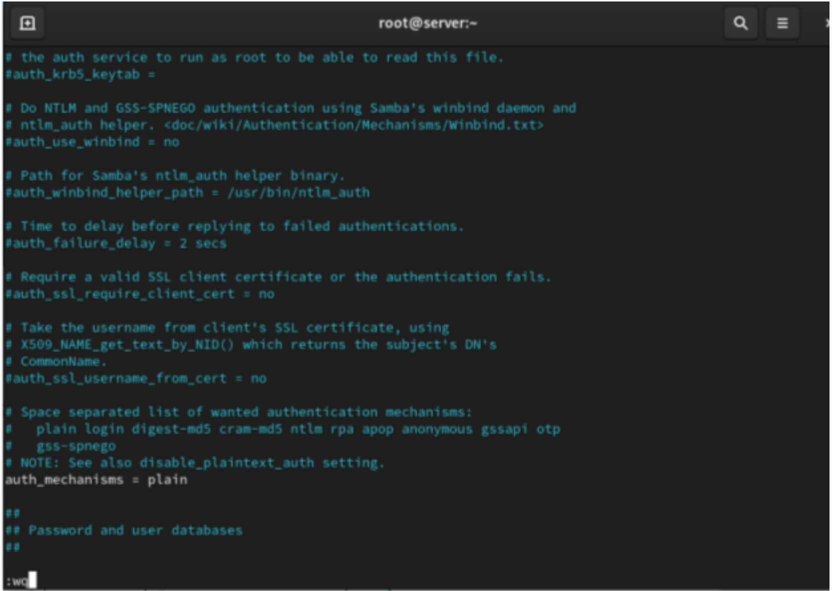
```
Downloading Packages:
(1/4): libexttextcat-3.4.5-11.el9.x86_64.rpm     40 kB/s | 209 kB     00:05
(2/4): clucene-core-2.3.3.4-42.20130812.e8e3d20 109 kB/s | 585 kB     00:05
(3/4): telnet-0.17-85.el9.x86_64.rpm             12 kB/s |  63 kB     00:05
(4/4): dovecot-2.3.16-15.el9.x86_64.rpm          3.2 MB/s | 4.7 MB     00:01
--------------------------------------------------------------------------------
Total                                            452 kB/s | 5.6 MB     00:12
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                        1/1
  Installing       : clucene-core-2.3.3.4-42.20130812.e8e3d20git.el9.x86_64 1/4
  Installing       : libexttextcat-3.4.5-11.el9.x86_64                      2/4
  Running scriptlet: dovecot-1:2.3.16-15.el9.x86_64                         3/4
  Installing       : dovecot-1:2.3.16-15.el9.x86_64                         3/4
  Running scriptlet: dovecot-1:2.3.16-15.el9.x86_64                         3/4
  Installing       : telnet-1:0.17-85.el9.x86_64                            4/4
  Running scriptlet: dovecot-1:2.3.16-15.el9.x86_64                         4/4
  Running scriptlet: telnet-1:0.17-85.el9.x86_64                            4/4
  Verifying        : telnet-1:0.17-85.el9.x86_64                            1/4
  Verifying        : libexttextcat-3.4.5-11.el9.x86_64                      2/4
  Verifying        : clucene-core-2.3.3.4-42.20130812.e8e3d20git.el9.x86_64 3/4
  Verifying        : dovecot-1:2.3.16-15.el9.x86_64                         4/4

Installed:
  clucene-core-2.3.3.4-42.20130812.e8e3d20git.el9.x86_64     dovecot-1:2.3.16-15.el9.x86_64
  libexttextcat-3.4.5-11.el9.x86_64                          telnet-1:0.17-85.el9.x86_64

Complete!
[root@server ~]#
```

Рис. 2: Проверка механизма аутентификации в 10-auth.conf

Источники пользователей и паролей (PAM + passwd)

Рис. 3: Настройка источников аутентификации в auth-system.conf.ext

Расположение почтовых ящиков (Maildir)



Межсетевой экран и запуск служб

```
##
# Location for users' mailboxes. The default is empty, which means that Dovecot
# tries to find the mailboxes automatically. This won't work if the user
# doesn't yet have any mail, so you should explicitly tell Dovecot the full
# location.
#
# If you're using mbox, giving a path to the INBOX file (eg. /var/mail/%u)
# isn't enough. You'll also need to tell Dovecot where the other mailboxes are
# kept. This is called the "root mail directory", and it must be the first
# path given in the mail_location setting.
#
# There are a few special variables you can use, eg.:
#
#   %u - username
#   %n - user part in user@domain, same as %u if there's no domain
#   %d - domain part in user@domain, empty if there's no domain
#   %h - home directory
#
# See doc/wiki/Variables.txt for full list. Some examples:
#
#   mail_location = maildir:~/Maildir
#   mail_location = mbox:~/mail:INBOX=/var/mail/%u
#   mail_location = mbox:/var/mail/%d/%1n/%n:INDEX=/var/indexes/%d/%1n/%n
#
# <doc/wiki/MailLocation.txt>
#
mail_location = maildir:~/Maildir

# If you need to set multiple mailbox locations or want to change default
# namespace settings, you can do it by defining namespace sections.
#
# You can have private, shared and public namespaces. Private namespaces
# are for user's personal mails. Shared namespaces are for accessing other
# users' mailboxes that have been shared. Public namespaces are for shared
# mailboxes that are managed by sysadmin. If you create any shared or public
# namespaces you'll typically want to enable ACL plugin also, otherwise all
# users can access all the shared mailboxes, assuming they have permissions
# on filesystem level to do so.
namespace inbox {
```

Мониторинг работы почтовых служб

Рис. 6: Записи в журнале Postfix и Dovecot

Проверка почтового ящика на сервере

```
[root@server ~]# postconf -e 'home_mailbox = Maildir/'
[root@server ~]#
[root@server ~]#
[root@server ~]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit ausweisapp2
bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc bitcoin-testn
et bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit collectd cond
or-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-tls dock
er-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server finger foreman foreman-proxy fr
eeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia-client ganglia-master
git gpsd grafana gre high-availability http http3 https ident imap imaps ipfs ipp ipp-client ipsec irc ircs is
csi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop kshell kube-api kube-apiserver
kube-control-plane kube-control-plane-secure kube-controller-manager kube-controller-manager-secure kube-nodep
ort-services kube-scheduler kube-scheduler-secure kube-worker kubelet kubelet-readonly kubelet-worker ldap lda
ps libvirt libvirt-tls lightning-network llmnr llmnr-client llmnr-tcp llmnr-udp managesieve matrix mdns memcac
he minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula netbios-ns netdata-dashboar
d nfs nfs3 nmea-0183 nrpe ntp nut openvpn ovirt-imageio ovirt-storageconsole ovirt-vmconsole plex pmcd pmproxy
 pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-exporter proxy-dhcp ps2link ps3ne
tsrv ptp pulseaudio puppetmaster quassel radius rdp redis redis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt
-master samba samba-client samba-dc sane sip sips slp smtp smtp-submission smtps snmp snmptls snmptls-trap snm
ptrap spideroak-lansync spotify-sync squid ssdp ssh steam-streaming svdrp svn syncthing syncthing-gui syncthin
g-relay synergy syslog syslog-tls telnet tentacle tftp tile38 tinc tor-socks transmission-client upnp-client v
dsm vnc-server warpinator wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-discovery-tcp ws-
discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-server zerot
ier
[root@server ~]# firewall-cmd --add-service=pop3 --permanent
success
[root@server ~]# firewall-cmd --add-service=pop3s --permanent
success
[root@server ~]# firewall-cmd --add-service=imap --permanen
success
[root@server ~]# irewall-cmd --add-service=imaps --permanent
bash: irewall-cmd: command not found...
^[[A^[[B^[[B
^C
[root@server ~]# firewall-cmd --add-service=imaps --permanent
success
[root@server ~]# firewall-cmd --reload
success
[root@server ~]# firewall-cmd --list-services
cockpit dhcpv6-client imap imaps pop3 pop3s smtp ssh
[root@server ~]#
```

Evolution: идентификация пользователя

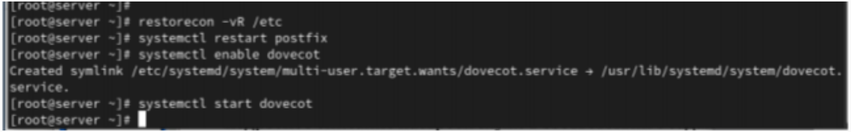


Рис. 8: Настройка учетной записи в Evolution — идентификация

Evolution: входящая почта (IMAP)

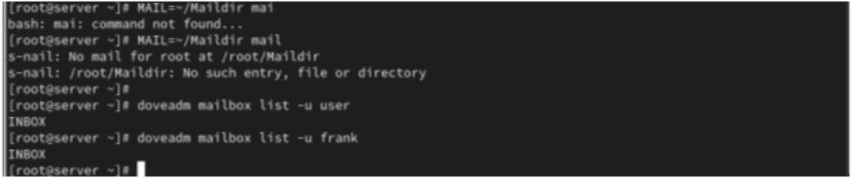
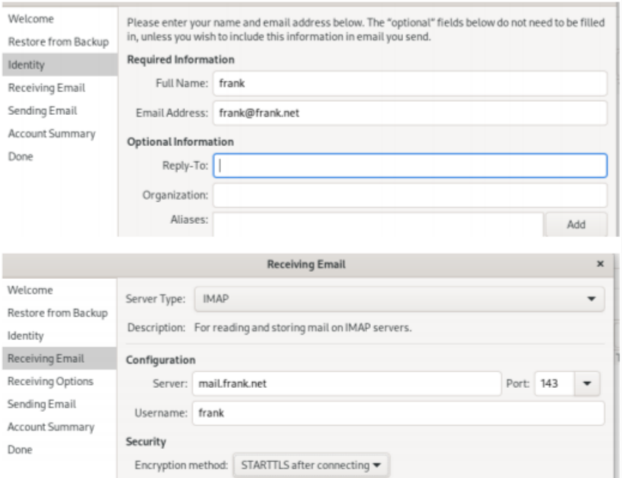


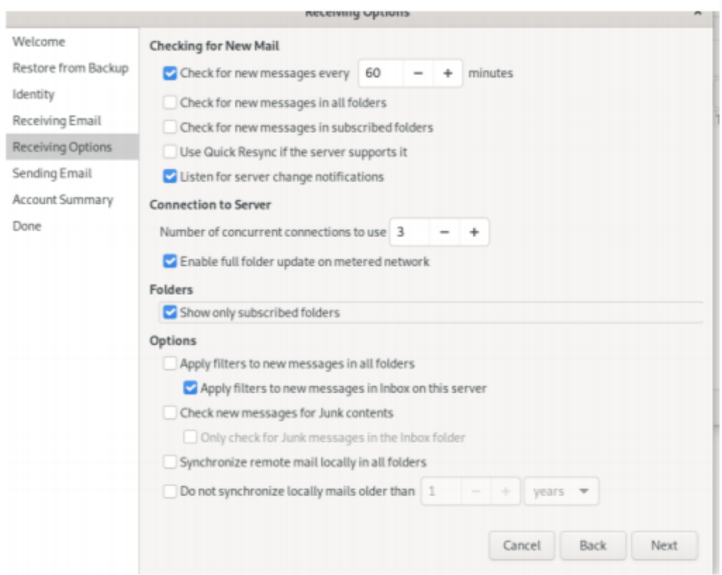
Рис. 9: Настройка входящей почты IMAP в Evolution

Evolution: исходящая почта (SMTP)

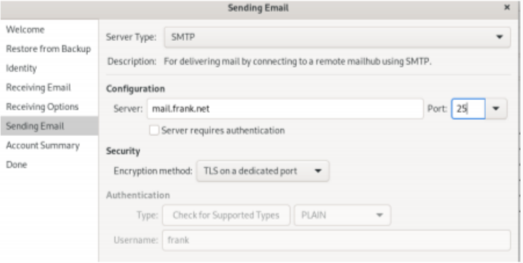
Проверены параметры SMTP: сервер mail.user.net, порт 25, без шифрования и без аутентификации.

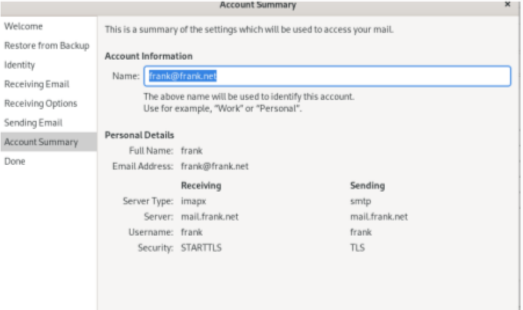


Отправка и получение тестовых писем



Проверка POP3 через Telnet





Выводы по проделанной работе

Вывод

В ходе лабораторной работы был установлен и настроен почтовый сервер на базе Postfix и Dovecot.
Выполнена настройка протоколов IMAP и POP3, обеспечено хранение почты в формате Maildir и корректная аутентификация пользователей.
Проверена работа почтовой системы с использованием Evolution, утилит mail/doveadm и проверки POP3 через Telnet.
Подтверждена корректная доставка, получение и удаление сообщений, а также готовность конфигурации к автоматическому развёртыванию в Vagrant.